

TECHNICAL BULLETIN

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**DODECENYLSUCCINIC ANHYDRIDE
AS AN EPOXY RESIN CURATIVE
(DDSA)
CAS Reg. No. 26544-38-7**

Typical Properties

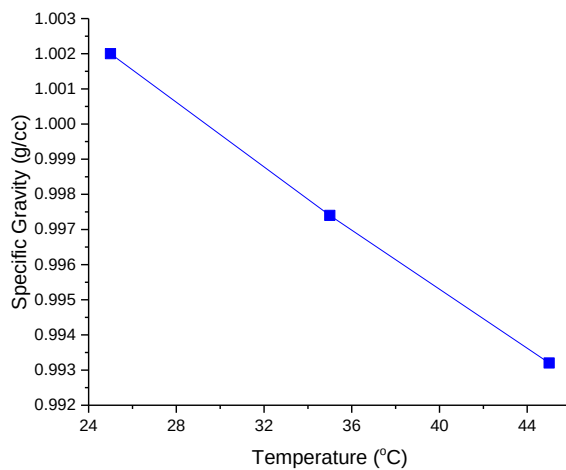
Properties	Value	Test Method
Molecular Weight (average)	266	
Specific gravity @ 25°C, g/cc	1.002	D-102 (ASTM D 4052)
Appearance	Light yellow liquid	D-104 (Visual)
Color, Gardner	6 max.	D-103
DDSA, %	97.0 min	D-126
Maleic Anhydride, %	0.2 max	D-126
Free Acid, %	2.0 max	D-126
Neutralization Equivalent, mg KOH/g	400-432	D-150i

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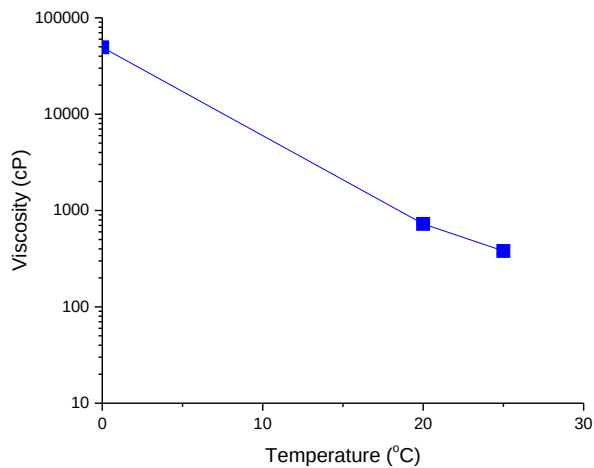
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Specific Gravity as a Function of Temperature



Viscosity as a Function of Temperature



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Applications

Dixie dodecenylsuccinic anhydride (DDSA) is a slightly viscous liquid anhydride. DDSA can be used as:

- An epoxy curing agent for electrical potting compounds, coatings, conductive adhesives, and syntactic foams
- An intermediate in the preparation of corrosion inhibitors for oils and greases
- As a leather tanning agent.

When used as an epoxy-curing agent, it has a long pot life at room temperature and gives excellent electrical properties and flexibility to castings. DDSA can be used in blends with other anhydrides to modify physical properties of the mixture and to lend flexibility and reduce brittleness of cured polymers.

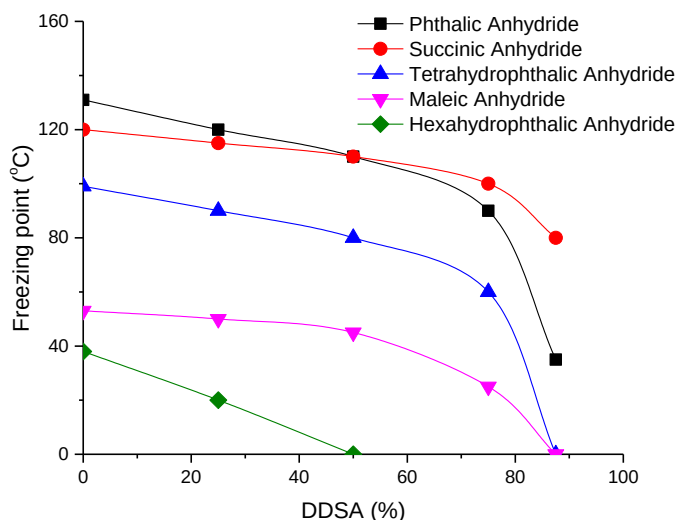
For more information on the use of anhydrides like DDSA as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical.

Modifying Other Anhydrides

As a liquid, DDSA is useful in modifying other anhydrides to lower their freezing points and make them easier to handle. The following plot shows how DDSA lowers the freezing point of five other anhydrides.

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Cure Catalysts

The effect of benzyldimethylamine (BDMA) catalyst on mix stability was studied using the following formulations:

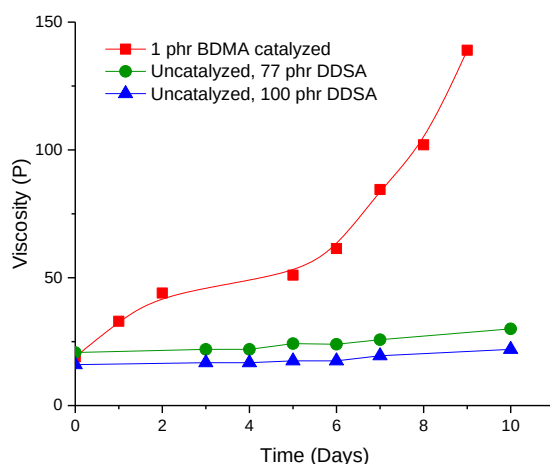
System	Catalyzed	Uncatalyzed	Uncatalyzed
Standard BPA Liquid Epoxy	100	100	100
DDSA	133	77	100
Benzyldimethylamine (BDMA)	1	0	0

* Epoxy homopolymerization discussed above is a significant side reaction for uncatalyzed systems, so anhydride demand is less.

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Viscosity as a function of time is summarized in the following plot:



Anhydride Level

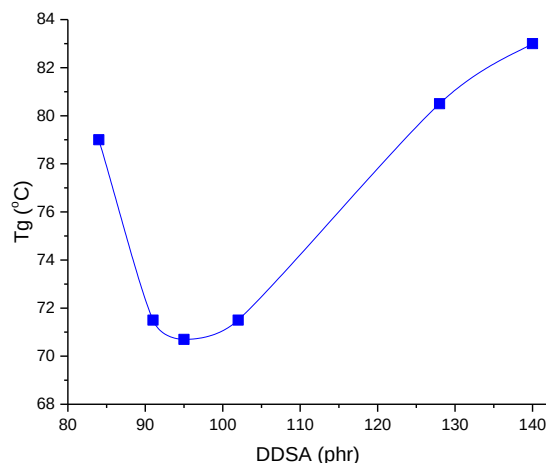
Most applications use between 80-150 phr DDSA with bisphenol A (BPA) liquid epoxy resins. Most high performance applications where high levels of heat or chemical resistance, or high strength is required utilize about 140 phr.

Example

Formulations with 100 parts BPA liquid epoxy resin, DDSA, and 1.5 part BDMA were prepared. The level of DDSA was varied from 84-140 phr. Samples were cured for 30 minutes at 100°C, the temperature was increased to 160°C in ten minutes, and then post-cured for 30 minutes. The following plot summarizes the glass transition temperatures (T_g) for these formulations:

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The same experiments were conducted with a 20:80 blend of DDSA and hexahydrophthalic anhydride. The following table summarizes the Tg data for these tests.

Anhydride blend, phr	Tg, °C
84	79
90	71
95	71
100	71
110	77
125	71
135	85
145	81

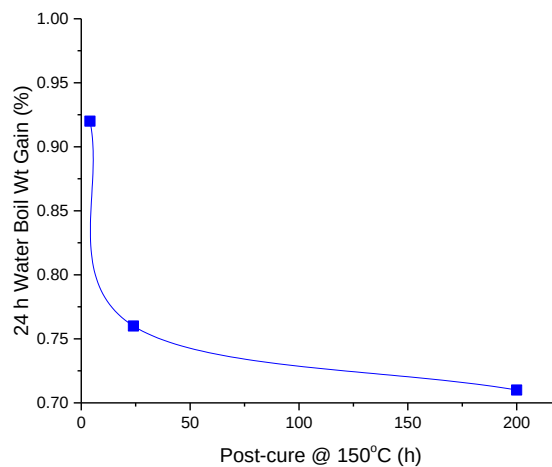
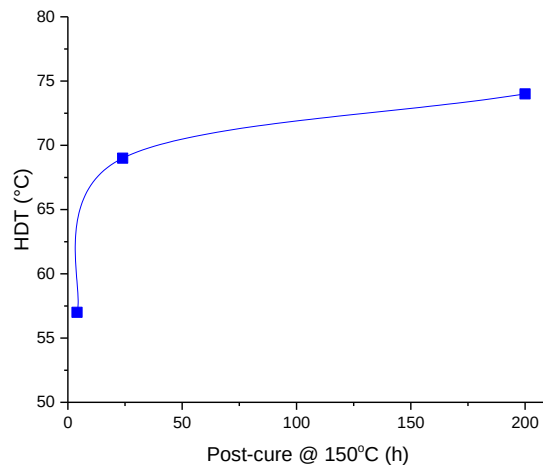
Cure Cycle

A formulation with 100 parts BPA liquid epoxy resin, 134 parts DDSA, and 1 part BDMA was prepared. Samples were cured for 4 hours at 90°C, and post-cured for 4, 24, and 200 hours at 150°C. The following plots show the

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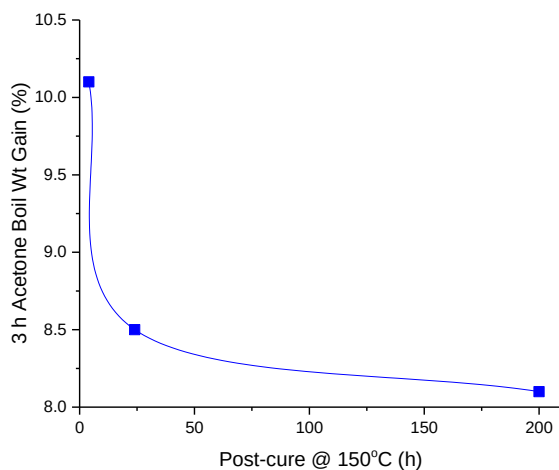
effect of post-cure on heat distortion temperature (HDT), and weight increases after 24 hours in boiling water and 3 hours in boiling acetone



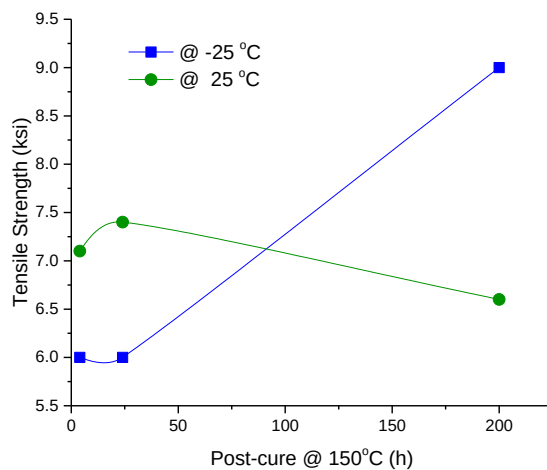
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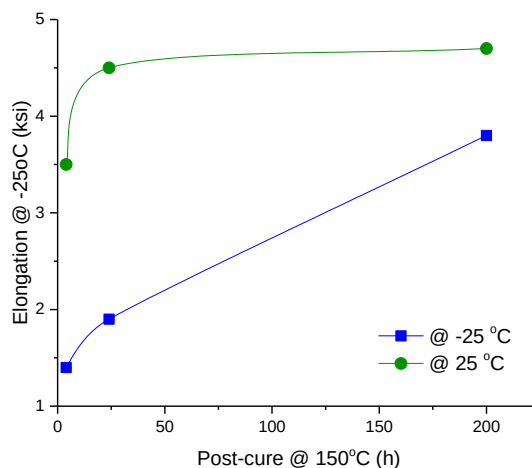
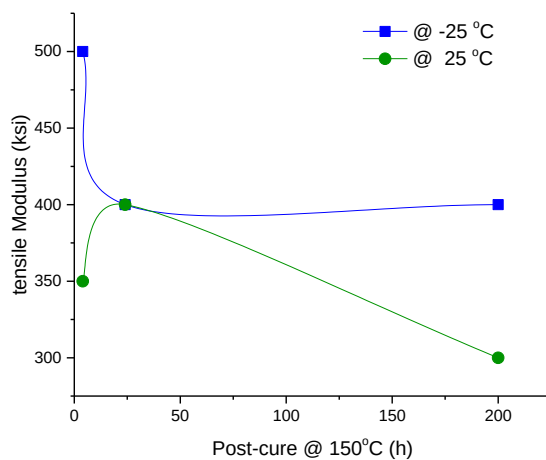


Tensile testing was performed at -25°C, and at room temperature. The following plots summarize results for tensile strength, tensile modulus, and elongation at break.



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Electrical Properties

A formulation with 100 parts BPA liquid epoxy resin, 138 parts DDSA, and 1.46 part BDMA was prepared. Samples were cured for 4 hours at 80°C, and post-cured for 24 hours at 100°C. The following table summarizes electrical, mechanical, and heat stability data for this formulation.

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Property	Condition	Value
Dielectric Constant	@ 60 Hz	2.85
	@ 1 MHz	2.79
Power Factor	@ 60 Hz	0.0039
	@ 1 MHz	0.0084
HDT, °C		77
Izod Impact, ft-lb/in. notch		0.22
Flexural Strength, ksi		13.2
Flexural Modulus, ksi		350
% Wt. loss 24 hr @ 200°C		0.35

Handling & Storage

DDSA will react with water to form dodecenylsuccinic acid. This is normally undesirable, so DDSA should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for DDSA, consult the Dixie Chemical brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

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Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

Dodecenylsuccinic anhydride is available from Dixie Chemical, Pasadena, Texas. Contact your Dixie representative for details.